

# Minimum central circles: certified finite optima and an effective global asymptotic constant

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## Abstract

Let  $R^*(n)$  be the least radius of a central circle to which nonoverlapping circles of radii  $1, \dots, n$  are externally tangent. We prove that  $R^*(n) = C_* n^2 + o(n^2)$  and characterize  $C_*$  by finite linear programs with an explicit error tending to zero. The reduction preserves arbitrary orders and all pairwise constraints: the limiting problem places marked points on a line at pairwise separation at least the geometric mean of their marks. Concatenation with a bounded boundary cost proves existence, and balanced finite-word programs supply matching effective upper and lower bounds. Their certified gap is  $(1/k + 1/r)/\pi$ , before directed arithmetic error, for  $k$  mark types and words of length  $r$ . We also transfer every finite or countable adjacent reflected-block construction whose total slab length is strictly below  $1 - \alpha$ , where  $0 \leq \alpha < 1$  is the shift parameter, to genuine permutations and full ring geometry, including a strict four-block improvement. The best retained explicit interval is  $C_{\text{term}} + \eta_{\text{width}} \leq C_* \leq U_4$ . The original computer-certified scope  $3 \leq n \leq 14$  and exact fixed-order Supnick feasibility theory remain distinct from global asymptotics. The earlier  $n^2/8$  conjecture is false. Efficient evaluation of the variational constant, an elementary expression and global floating-circle structure remain open. This candidate has undergone internal adversarial checks; independent external acceptance is pending.

## 1 Problem, scope and relation to the historical version

All outer circles are tangent to the central circle; outer circles need not be tangent to each other. For a cyclic order  $\sigma$ , the adjacent chain radius  $R_{\text{chain}}(\sigma)$  ignores nonadjacent constraints, whereas  $R_{\text{full}}(\sigma)$  is the least radius satisfying every pair constraint in that order. The global quantity is  $R^*(n) = \min_{\sigma} R_{\text{full}}(\sigma)$ . Always  $R_{\text{chain}}(\sigma) \leq R_{\text{full}}(\sigma)$ . A feasible construction supplies an upper bound, not an optimum certificate.

The public arXiv-v1 manuscript [1] reported certified finite optima and proposed the asymptotic coefficient  $1/8$ . That version is an immutable historical record. Post-v1 lower bounds disprove its asymptotic conjecture. The central contribution of this version is an effective characterization of the true leading coefficient, proved in Sections 2–4. It is independent of the longer proof chains used for the explicit lower and upper endpoints. Those endpoints and fixed-order results are retained with their exact definitions and source dependencies in the supplement.

The line problem has a familiar geometric interpretation. Disks of radii  $a_i$  tangent to the horizontal axis, with centers  $(2x_i, a_i)$ , are nonoverlapping precisely when  $|x_i - x_j| \geq \sqrt{a_i a_j}$ . For  $a_i \leq 1$ , their horizontal envelope span differs from  $2(\max_i x_i - \min_i x_i)$  by at most two. Minimum-span disks on a shelf are studied by Alt et al. [2]. We do not claim this underlying model as new; our asymptotic reduction, uniform-label recovery and effective balanced-word characterization

are proved directly below, without importing an approximation or complexity theorem for that model.

## 2 Exact angular comparison

The cosine law gives the required pair angle

$$\theta_R(u, v) = 2 \arcsin \sqrt{\frac{uv}{(R+u)(R+v)}} = 2 \arctan \sqrt{\frac{uv}{R(R+u+v)}}. \quad (1)$$

Both directed cyclic arcs between a pair must be at least this angle. This is equivalent to non-overlap, since their smaller separation belongs to  $[0, \pi]$ . The angle is positive, symmetric, increasing in either outer radius and strictly decreasing in  $R > 0$ .

**Definition 1.** For a finite multiset  $a = (a_1, \dots, a_N)$  of marks in  $[0, 1]$ , let  $b(a)$  be the least span of a line placement satisfying

$$|x_i - x_j| \geq \sqrt{a_i a_j} \quad (i \neq j). \quad (2)$$

Distinct zero-mark points may coincide. Write  $b_n = b(1/n, 2/n, \dots, 1)$ .

For a fixed word  $w = (w_1, \dots, w_N)$  in left-to-right order, its minimum span  $\ell(w)$  is attained by

$$x_1 = 0, \quad x_i = \max_{j < i} \{x_j + \sqrt{w_i w_j}\}, \quad \ell(w) = x_N. \quad (3)$$

Induction proves that these coordinates are componentwise no larger than any feasible ordered coordinates with  $x_1 = 0$ ; they also satisfy every constraint. Nonnegative weights ensure ordered coordinates. Equivalently,  $\ell(w)$  is the longest increasing-index path from 1 to  $N$ , with edge weight  $\sqrt{w_i w_j}$ . Thus  $b(a)$  is an attained minimum over a finite set of orders. Deleting or decreasing marks cannot increase it; scaling all marks by  $h$  scales the minimum span by  $h$ .

**Lemma 2** (Bounded closing cost). *Let  $B(a)$  be the infimum circumference with both pair arcs at least  $\sqrt{a_i a_j}$ . Then  $b(a) \leq B(a) \leq b(a) + 1$ .*

*Proof.* Cut a feasible circle to obtain a feasible line placement. Conversely, close a feasible line placement by an additional gap of length one. One arc for every pair is its line separation; the other contains the added gap and is at least  $1 \geq \sqrt{a_i a_j}$ . Both arcs are controlled.  $\square$

**Lemma 3** (Exact geometric squeeze). *For every  $n \geq 3$ ,*

$$\frac{b_n}{\pi n} - \frac{1}{n} \leq \frac{R^*(n)}{n^2} \leq \frac{b_n + 1}{\pi n}. \quad (4)$$

*Proof.* For  $1 \leq u, v \leq n$ , the two expressions in (1) give

$$\frac{2\sqrt{uv}}{R+n} \leq \theta_R(u, v) \leq \frac{2\sqrt{uv}}{R}.$$

The first uses  $\arcsin z \geq z$  and  $\sqrt{(R+u)(R+v)} \leq R+n$ ; the second uses  $\arctan z \leq z$ . Scaling a feasible ring's angles by  $(R+n)/(2n)$  gives a feasible linear circle of circumference  $\pi(R+n)/n$ , proving the lower inequality. Conversely, the circumference  $b_n + 1$  from Lemma 2, scaled to  $2\pi$ , satisfies every original pair angle at  $R = n(b_n + 1)/\pi$ . This proves the upper inequality. No Taylor expansion, adjacency reduction or assumption on the optimal order was used.  $\square$

### 3 Existence by genuine-label concatenation

**Theorem 4** (The leading limit exists). *There is a constant  $C_*$  such that*

$$R^*(n) = C_* n^2 + o(n^2), \quad \pi C_* = E := \lim_{n \rightarrow \infty} \frac{b_n}{n} = \inf_{n \geq 2} \frac{b_n + 1}{n}. \quad (5)$$

*Proof.* Copies of a feasible line template, separated by unit gaps, satisfy all cross-copy constraints as well as the internal ones. Fix  $n \geq 2$ . Given  $N \geq n$ , let  $q = \lceil N/n \rceil$  and concatenate  $q$  copies of an optimal  $b_n$  template. Its span is  $q(b_n + 1) - 1$ . Retain the  $N$  smallest marks. The  $i$ th sorted retained mark is  $\lceil i/q \rceil/n$ . Scale all marks and coordinates by  $h = nq/N$ . The resulting  $i$ th mark is at least  $i/N$ , so decreasing it to  $i/N$  preserves every constraint. Each target label is assigned exactly once to an actual retained point, with ties broken arbitrarily. Hence

$$b_N \leq \frac{nq}{N} [q(b_n + 1) - 1]. \quad (6)$$

Intermediate marks exceeding one cause no problem: the line inequalities are homogeneous, and final marks are exactly  $1/N, \dots, 1$ .

As  $N \rightarrow \infty$ ,  $q/N \rightarrow 1/n$ , giving  $\limsup_N b_N/N \leq (b_n + 1)/n$  for every fixed  $n$ . The sequence is bounded between zero and one by equally spaced placements. Taking  $n$  along a subsequence attaining its liminf gives equality of limsup and liminf. Taking the infimum and then  $n \rightarrow \infty$  proves the last equality in (5). Lemma 3 transfers the limit to the original ring problem for all sizes and both parities. Finally, at least  $n/2$  marks are at least  $1/2$ . Their successive line separations are at least  $1/2$ , so  $E \geq 1/4$  and  $C_* > 0$ .  $\square$

## 4 Effective finite-word characterization

### 4.1 Finite types and a uniform quantization error

For  $k \geq 1$ , let  $A_{k,q}$  contain  $q$  copies of each mark  $1/k, \dots, 1$ , and let  $A_{k,q}^-$  instead contain  $q$  copies of  $0, 1/k, \dots, (k-1)/k$ . Define  $e_k = \lim_{q \rightarrow \infty} b(A_{k,q})/(kq)$  and  $e_k^- = \lim_{q \rightarrow \infty} b(A_{k,q}^-)/(kq)$ ; these limits exist. Indeed  $a_q = b(A_{k,q}) + 1$  is subadditive by unit-gap concatenation. Writing  $q = hp + s$  at fixed  $p$  proves  $\limsup a_q/q \leq a_p/p$ ; then taking  $p$  along a liminf subsequence proves the limit. The same argument applies to the lower types.

The two multisets share every nonzero type except the top type. Zero marks cost nothing. Adding the  $q$  unit marks as a separate equally spaced block costs at most  $q$ , including the join. Thus, also for  $k = 1$ ,

$$e_k^- \leq e_k \leq e_k^- + \frac{1}{k}.$$

For  $n = kq$ , the sorted uniform marks  $i/n$  lie between the two sorted type multisets. Theorem 4 and monotonicity imply

$$e_k - \frac{1}{k} \leq E \leq e_k. \quad (7)$$

## 4.2 Balanced words and the quantified boundary relaxation

Fix  $k \geq 1$  and  $r \geq 2$ , and use all  $k^r$  words  $w \in \{1/k, \dots, 1\}^r$ , including repetitions. Let  $c_j(w)$  count occurrences of  $j/k$ . Compute  $\ell(w)$  by (3), and define

$$\begin{aligned} \lambda_{k,r} &= \min_p \frac{1}{r} \sum_w p_w \ell(w), \\ p_w &\geq 0, \quad \sum_w p_w = 1, \quad \sum_w p_w c_j(w) = \frac{r}{k} \quad (1 \leq j \leq k). \end{aligned} \tag{8}$$

The polytope is nonempty: equally mix the  $k$  constant words.

**Theorem 5** (Matching effective variational bounds). *For every  $k \geq 1, r \geq 2$ ,*

$$\boxed{\frac{\lambda_{k,r} - 1/k}{\pi} \leq C_* \leq \frac{\lambda_{k,r} + 1/r}{\pi}}. \tag{9}$$

Consequently  $C_* = \pi^{-1} \lim_{k,r \rightarrow \infty} \lambda_{k,r}$ , jointly, with certified bracket width  $(1/k + 1/r)/\pi$ .

*Proof.* Partition an optimal ordering of  $A_{k,q}$  into consecutive blocks of  $r$  vertices, discarding fewer than  $r$  leftover vertices. Their spans sum to at most the full span and each is at least its word cost. Empirical word distributions have convergent subsequences in the finite simplex. Their limiting mean type counts are  $r/k$ , since the discarded counts vanish after normalization. Hence  $\lambda_{k,r} \leq e_k$ .

Conversely, the rational feasible polytope in (8) has a minimizing rational vertex even though its cost entries are algebraic. Choose a common denominator  $D$ . A batch with  $Dp_w$  copies of each word has exactly  $Dr/k$  copies of each type, an integer by the count constraints. Place every word at its minimum span and separate blocks by unit gaps. All within-block and cross-block pairs are feasible. Repeating batches gives asymptotic mean span  $\sum_w p_w (\ell(w) + 1)/r = \lambda_{k,r} + 1/r$ . The finite-type limit therefore satisfies  $e_k \leq \lambda_{k,r} + 1/r$ . Combine with (7).  $\square$

## 4.3 Exact certificates and computational meaning

Directed rational square-root enclosures of width  $\epsilon$  give word-span enclosures of width at most  $(r-1)\epsilon$ : every path in (3) has at most  $r-1$  edges. Rational endpoint LPs then enclose  $\lambda_{k,r}$  with additional width at most  $(r-1)\epsilon/r$ . They can be solved by finite exact vertex enumeration. Practical solvers may instead discover candidates, followed by exact checks. For a rational lower cost  $\ell^-(w)$ , dual coefficients satisfying

$$y_0 + \sum_j y_j c_j(w) \leq \ell^-(w)/r \quad \text{for every word}$$

certify the lower value  $y_0 + (r/k) \sum_j y_j$ . Feasible rational  $p$  certifies an upper value using upper costs. All such checks are finite rational comparisons; equality is not decided by indefinite interval refinement. Together with a rational enclosure of  $\pi$ , this proves arbitrary-precision computability of  $C_*$  in principle.

The word count is exponential in  $r$ . No claim of cheap high-precision evaluation or a closed-form constant follows. The bounded implementation uses SciPy only to propose sparse primal and dual candidates; exact Fraction arithmetic validates every equality and word inequality. Small word programs need not improve the stronger explicit endpoint bounds below.

## 5 Explicit geometric upper constructions

Fix  $0 \leq \alpha < 1$  and nonnegative adjacent slab lengths  $\ell_1, \dots, \ell_k$  of total  $T < 1 - \alpha$ . Put  $A = 1 + \alpha$ ,  $t_0 = 0$ ,  $t_j = \sum_{h \leq j} \ell_h$ . On each slab  $[t_{j-1}, t_j]$ , use equal masses at

$$(t, A + t, A + t_{j-1} + t_j - t), \quad (t, A + t_{j-1} + t_j - t, A + t).$$

After  $T$ , use the diagonal  $(t, 1 + \{t + \alpha\}, 1 + \{t + \alpha\})$ . This defines a measure  $\mu$  on  $[0, 1] \times [1, 2]^2$  with uniform low and separate high marginals. Its complete cell cost is

$$C(\mu) = \frac{1}{4\pi} \int \max\{\sqrt{t}(\sqrt{X} + \sqrt{Y}), \sqrt{XY}\} d\mu. \quad (10)$$

Correct marginals alone do not establish finite recovery.

**Theorem 6** (General block transfer). *Every such finite partition has genuine alternating-half permutations with all-pairs-feasible radii  $\rho_m = R_{\text{full}}(\sigma_m)$  and*

$$\left| \frac{\rho_m}{(2m)^2} - C(\mu) \right| \leq \frac{(6k^2 + 28k + 1060)/m + 16384/(3m^2)}{4\pi}$$

for  $m \geq \max\{2048, \lceil 2/(1 - \alpha - T) \rceil\}$ . It follows that  $C_* \leq C(\mu)$ . Countably many slabs also transfer if their total length is strictly below  $1 - \alpha$ .

*Proof with explicit construction.* Let  $s = \lfloor \alpha m \rfloor$ ,  $l_j = 2\lfloor \ell_j m/2 \rfloor$  and  $a_j = \sum_{h \leq j} l_h$ . Reverse only the even ranks within each  $(a_{j-1}, a_j]$ , fixing odd ranks, to obtain a permutation  $J$ . Set  $P_i = m + 1 + ((J(i) + s - 1) \bmod m)$  and  $\sigma_m = (1, P_1, \dots, m, P_m)$ , with  $P_0 = P_m$ . Disjoint involutions and the cyclic shift prove bijectivity, including zero blocks and length-two identities. The integer gate keeps the final block endpoint at least two ranks before the high wrap, and its exit strictly before that wrap.

Double panels of width  $2/m$ , allocated half to each parity, recover the two reflected orientations. Interior coordinate error is at most  $4/m$ . There are at most  $k + 3$  exceptional actual cells: shared exits, the cyclic predecessor, and the two wrap cells. Rounded slab boundaries and wrap differ on a set of length at most  $[k(k + 1) + 1]/m$ , with remaining coordinate error at most  $(4k - 1)/m$ . Write  $g(t, X, Y) = \max\{\sqrt{t}(\sqrt{X} + \sqrt{Y}), \sqrt{XY}\}$  for the integrand in (10). It is 4-Lipschitz and lies in  $[1, 3]$ . The actual normalized full-cell moment is

$$G_m = \frac{1}{m} \sum_{i=1}^m g(i/m, P_{i-1}/m, P_i/m).$$

It differs from  $\int g d\mu$  by at most  $(6k^2 + 28k + 36)/m$ .

For arbitrary distinct highs in  $[m + 1, 2m]$ , the exact full-feasibility criterion for this alternating order is

$$\sum_{i=1}^m \max\{\theta_R(P_{i-1}, i) + \theta_R(i, P_i), \theta_R(P_{i-1}, P_i)\} \leq 2\pi.$$

Its sufficiency follows from the high-shell triangle inequality  $2\theta_R(m + 1, m + 1) > \theta_R(2m, 2m)$  and high/low separation: each directed high path contracts through whole cells, and paths with low endpoints either contain such a high path or an incident high gap that already exceeds the endpoint requirement. Both directions are treated separately. At its unique root, choose the first

half of each cell at its required low/high angle and put any excess in the other half. This closes at  $2\pi$  and satisfies all pairs.

At  $R = 4cm^2$ ,  $c \geq 1/32$ , the full score differs from  $G_m/(2c)$  by at most  $1024/m + 16384/(3m^2)$ . For  $m \geq 2048$  this first brackets the actual normalized root in  $(1/32, 1/2)$ , then yields the stated error. Deleting label  $2m$  gives the odd upper bound. For countably many slabs, truncate and use that the omitted complete-cost integral is at most six times the omitted length. A diagonal sequence with  $m$  growing faster than  $k^3$  pays both finite and tail errors. Full cell/seam formulas and the angle estimate are supplied in the versioned proof supplement [5].  $\square$

For the retained explicit construction, let  $x_*$  be the unique minimizer on  $[0, 1]$  of

$$E(x) = \int_0^x \left[ \max\{\sqrt{(1+t)(1+x-t)}, \sqrt{t}(\sqrt{1+t} + \sqrt{1+x-t})\} - \max\{1+t, 2\sqrt{t(1+t)}\} \right] dt.$$

Let  $K(\alpha) = \int_0^1 \max\{\sqrt{th_\alpha(t)}, h_\alpha(t)/2\} dt$ , where  $h_\alpha(t) = 1 + \{t + \alpha\}$ , and let  $\hat{\alpha}$  be the unique zero of  $K'(\alpha) + (1 + \alpha)E(x_*)$  on  $[0, 1/2]$ . Set  $A = 1 + \hat{\alpha}$  and  $\lambda = Ax_*$ . The second width  $\epsilon_b$  is the unique mixed-branch stationary minimum for a separate reflection starting at  $\lambda$ ; the third width  $\Delta_*$  is the analogous unique minimum starting at  $v = \lambda + \epsilon_b$ . Each width is minimized over  $[0, A/3 - u]$  at its fixed start  $u$ , retaining the full max. The supplement [6] gives their exact defining integrals, existence/uniqueness proofs and rational isolations:

$$\begin{aligned} .1093 < \hat{\alpha} < .10931, & \quad .2876 < x_* < .2877, \\ .043 < \epsilon_b < 7/160, & \quad 29/5000 < \Delta_* < 27/4000. \end{aligned}$$

These terminating decimals are rational bracket endpoints, not definitions or numerical replacements for the parameters.

Let  $C_3(d)$  be (10) for the three lengths  $(\lambda, \epsilon_b, d)$ , and  $C_4(\eta)$  the same cost for the four lengths  $(\lambda, \epsilon_b, \Delta_*, \eta)$ . Use  $\eta = 1/20000$  and define  $U_4 = C_4(1/20000)$ . If  $w = \lambda + \epsilon_b + \Delta_*$ ,  $B = A + w$  and  $M = B + \eta/2$ , the fourth increment at  $0 < \eta \leq 1/20000$  is exactly

$$C_4(\eta) - C_3(\Delta_*) = -\frac{1}{4\pi} \int_{-\eta/2}^{\eta/2} \frac{z^2}{M + \sqrt{M^2 - z^2}} dz.$$

Rational input margins keep both changed maxima strictly chord. Since  $2B \leq M + \sqrt{M^2 - z^2} \leq 2M$ ,  $M < 3/2$  and  $\pi < 4$ ,

$$C_* \leq U_4 < C_3(\Delta_*) - \frac{1}{4608000000000000}. \quad (11)$$

The leading decrement is  $-\eta^3/(96\pi B)$ , with remainder between zero and  $\eta^4/(192\pi B^2)$ . The theorem transfers this literal continuous saving to genuine full geometry. It does not make four blocks globally optimal or optimize an infinite partition.

## 6 The strongest retained explicit lower endpoint

Define  $\tau$  as the unique root of  $\cos \tau = \tau$  in  $(0, 1)$ , set  $q = (1 - \sin \tau)/(1 + \sin \tau)$  and  $C_{\text{term}} = \tau(1 + q)/(2\pi)$ . For the fixed cutoffs

$$(\beta_1, \beta_2, \beta_3) = (2091/10000, 10907/50000, 23/100)$$

put

$$D_i = \int_q^{\beta_i} [1 + q - x - 2\sqrt{x(1 + q - x)}] dx, \quad F(h) = \sum_i h_i D_i - 8 \sum_i h_i^3.$$

Let  $a = 113/12500$ ,  $b = 593/50000$ . On  $h_i \geq 0$ ,  $h_1 + h_2 \leq a$ ,  $h_2 + h_3 \leq b$ , define

$$\eta_{\text{width}} = \max_h \frac{\max\{F(h), 0\}}{\pi(16 + 432 \sum_i h_i)}.$$

This maximum is uniquely attained at  $(a - z_*, z_*, b - z_*)$ , where  $z_*$  is the unique zero on  $[0, a]$  of  $G = N'L + 432N$ ,  $N(z) = F(a - z, z, b - z)$  and  $L(z) = 16 + 432(a + b - z)$ .

**Proposition 7** (Explicit lower bound, post-v1). *The global constant satisfies  $C_* \geq C_{\text{term}} + \eta_{\text{width}}$ , with exact directed enclosure*

$$.14056946887766098063257 < C_{\text{term}} + \eta_{\text{width}} < .14056946887766098063392.$$

*In particular  $C_* > 1/8$ , so both the v1 coefficient conjecture and its stronger  $n^2/8 - R^*(n) = O(\sqrt{n})$  claim are false.*

*Dependency-explicit proof outline.* The complete proof and exact arithmetic certificates are in [7]. Its essential coupled information is one shared crossing energy for the same order and all three nested terminal restrictions. An explicit assignment dual bounds the reflected-grid energy by eight times the excess chain cost. The absolute signed deletion discrepancy at cutoff  $i$  is at most 54 times this energy plus twice its crossing term. Separated strips charge the common energy once, yielding the scalar gain  $\max\{F(h), 0\}/(16 + 432 \sum h_i)$ . The exact angular squeeze transfers each induced chain bound to any original full feasible configuration. Uniform finite errors vanish at every fixed strictly separated width vector.

For the width optimum,  $G' < 0$  and rational endpoint gates isolate its zero. Set  $u = (a - z_*, z_*, b - z_*)$ ,  $\mathcal{L}(h) = 16 + 432 \sum_i h_i$  and  $\rho = F(u)/\mathcal{L}(u)$ . The gradient of  $F - \rho\mathcal{L}$  at  $u$  is a positive combination of the two constraint normals. Its exact remainder is  $-8 \sum_i (h_i - u_i)^2 (h_i + 2u_i)$ , proving global uniqueness on the whole closed region. To pass to the geometric limit, first use  $tu$  at fixed  $0 < t < 1$ , take  $n \rightarrow \infty$ , then  $t \uparrow 1$ . The unchanged boundary widths fail finite floor separation infinitely often; the proof never uses them as an eventual finite gate. The separate internal lower audit in [7] rederives this minimal dependency path. The result is not a claim that this particular lower mechanism reaches  $C_*$ .  $\square$

## 7 Fixed-order theory and finite certification

For increasing positive radii, the positive mixed derivative of  $\theta_R(a, b)$  makes its matrix anti-Monge. Applying the classical maximum Supnick-tour theorem to its negative selects one minimizing chain order, independently of  $R$  [3, 4]. This proves chain optimality, not full feasibility after a seam obstruction.

For integers  $k \geq 1$  and  $n \geq k + 2$ , consider the canonical order on  $\{k, \dots, n\}$ . Write  $R_{k,n}$  for its chain root and  $\delta_{k,n} = \theta_R(n, k) + \theta_R(k, n - 1) - \theta_R(n, n - 1)$  at that root. The exact post-v1 all-pairs classification [8] is:

| $k$        | Feasible at the chain root | Infeasible there |
|------------|----------------------------|------------------|
| 1          | $3 \leq n \leq 7$          | $n \geq 8$       |
| 2          | $4 \leq n \leq 12$         | $n \geq 13$      |
| 3          | $5 \leq n \leq 16$         | $n \geq 17$      |
| 4          | $6 \leq n \leq 20$         | $n \geq 21$      |
| 5          | $7 \leq n \leq 24$         | $n \geq 25$      |
| $k \geq 6$ | $k + 2 \leq n \leq 4k + 5$ | $n \geq 4k + 6$  |

The underlying iff criterion is  $\delta_{k,n} \geq 0$ : its seam is the minimum triangle defect, and fan telescoping controls both arcs for every pair. At the chain root every adjacent gap is forced tight, so a negative seam cannot be repaired there. The integer seam signs are strict. None of this determines the global floating set or the full optimum on the larger set after selected radii are deleted.

The original finite certificates report global optima for every  $3 \leq n \leq 14$ , with absolute radius tolerance  $10^{-10}$  and local bracket scale  $10^{-12}$ . The independent verifier reconstructs all-pair and Cartesian feasibility at 50 decimal digits, checks the local brackets, and audits the complete saved pruning frontiers and their provenance. This candidate does not expand that finite scope or regenerate the exhaustive searches. These numerical certificates retain the original floating-point error guards; verification checks the saved frontier and coverage summaries, not every excluded order with directed interval arithmetic. The full current local run passes at all twelve sizes, including the 3,113,510,400 canonical orders covered at  $n = 14$ . Larger- $n$  reported candidates remain heuristic upper bounds subject to feasibility checks, not certified optima.

## 8 Reproducibility, review status and remaining questions

The repository supplement is fixed at commit `3beb8d70c5b3748d370a92855847bdf574e5a14f` [9]; all supplementary paths below refer to that snapshot unless an earlier checkpoint is explicitly specified. It contains complete mathematical proofs, exact parameter checkers, independent arithmetic controls and task-specific internal audits. New line/word checks include omitted non-adjacent constraints, absent closing gaps, invalid label recovery, corrupt primal/dual certificates and disabled-assertion runs. Their bounded computations corroborate finite cores; the analytic proofs establish infinite quantifiers.

The original progress logs are preserved byte for byte in deterministic gzip archives with original/archive hashes. Restore them before the full verifier:

```
python scripts/frontier_logs.py restore
python -m pytest
python verify.py --start 3 --stop 8 --skip-frontier
python verify.py --start 3 --stop 14
```

The smoke command skips global frontiers and must not be called a global certificate check. Exact new checker commands, environment and clean-source reproduction evidence are listed in the review packet [9]. No hosted CI status or external acceptance is inferred from local success.

The leading normalized behavior is now characterized effectively. Remaining questions concern practical evaluation or simplification of  $C_*$ , sharp explicit constructions, microscopic optimizer structure, subleading terms, and the global floating set. The current study neither claims an elementary closed form nor infers contact structure from the variational limit. AI assistance was used for proof exploration, software, drafting and internal adversarial checks. These checks remain distinct from the author’s final approval and independent external review.



## References

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- [5] Ringmin proof supplement, `research/PERMUTED_HALVES_GENERAL_BLOCK_TRANSFER.md` and `research/PERMUTED_HALVES_FOURTH_ADJACENT_BLOCK.md`, mathematical checkpoint 13ddb41180b3911940f4fe5cf7d61c0545f9f834.
- [6] Ringmin supplement, `ops/GOAL-20260911__conclude_study/UPPER_INTERNAL_AUDIT.md`, with the four linked exact parameter proofs, at the same mathematical checkpoint.
- [7] Ringmin supplement, `research/THREE_LEVEL_COMMON_CHAIN.md`, Sections 7, 9, 11–12; `ops/GOAL-20260911__conclude_study/LOWER_INTERNAL_AUDIT.md`, with the minimal proof dependency graph and exact checks, at that checkpoint.
- [8] Ringmin supplement, `research/SUPNICK_FULL_FEASIBILITY.md` and `research/SUPNICK_SEAM_SEQUENCES.md`; final dependency audit in `ops/GOAL-20260911__conclude_study/FIXED_ORDER_INTERNAL_AUDIT.md`.
- [9] Ringmin final review packet and source manifest, `ops/GOAL-20260911__conclude_study/FINAL_REVIEW_PACKET.md`. Repository: <https://github.com/falker47/ringmin>. Supplementary snapshot: repository at the reviewed commit. Internal validation and independent external acceptance are explicitly separated there.